

frontmatter — decisions

What was decided across twelve research rounds, what later evidence overturned, and where all 105 reports live.

v1.0 · 2026-08-29 · owner: Sagnik Mitra The third of three. PRD = what and why. BUILD-PLAN = in what order. This = what was decided, what was found to be wrong, and where the evidence lives.

Contents

- 00 What this is
- 01 Decisions now made
 - 1.1 Architecture · 1.2 Why not a CRDT — the three disqualifications · 1.3 Product and commercial · 1.4 Operational realities that constrain the build

- 02 Corrections to the PRD
- 03 Still open — and each one re-scopes a lane
- 04 Research index
- 05 What we may say in public, and what we may not

§0 What this is

Twelve research rounds, **105 reports, 374,866 words**. This document is the accumulation: every decision that is now made and why, every claim in the PRD that later evidence overturned, what remains genuinely open, and an index so any of it can be traced back.

The rule that governs the whole record: a decision here is only as good as its falsifier. Every row carries "what would change this". A decision with no falsifier is a preference.

§1 Decisions now made

1.1 Architecture

#	DECISION	REASONING	WHAT WOULD CHANGE IT
A1	Sync = git three-way merge + an append-only splice journal, server-mediated by compare-and-swap. Not CRDT, not OT, not fuzzy patch	Three independent disqualifications, all measured or fetched — see §1.2	A measured conflict rate above the published budget. The fix would be finer merge granularity, never a fuzzier apply
A1b	T0's exit condition is "zero bytes lost, every divergence surfaced as a reviewable hunk" — not "convergence"	"Convergence" is a CRDT word, and CRDTs converge to interleaved garbage (§1.2). The architecture that must satisfy T0 is merge3 + OCC	—
A2	Git repo = document truth, never moves. One Postgres as control plane holding zero document bytes. R2 for attachments >1 MB and derived artifacts	Publish-slug uniqueness currently scans the whole snapshot; identity, entitlements, billing and audit have no home	D8 resolving to local-first/E2E shrinks the control plane to identity + billing
A3	<code>workspace_id</code> on every row from day one, pooled RLS. The GitHub App installation is the connector , never the tenant identity	Users connect two repos, transfer repos, uninstall. Identity must survive all three. Retrofitting <code>workspace_id</code> post-launch is the highest-cost change on the board	An enterprise deal needing infrastructure-level isolation — then a dedicated tier, priced separately
A5	Stop shipping the whole snapshot → Postgres FTS + trigram server-side → MiniSearch client-side only. CJK via a bigram tokenizer , not a dictionary segmenter	77.5 MB parsed per cold start; the snapshot response is 16.4× the platform body cap	p95 above 400 ms on Postgres FTS at real corpus size
A14	MDMAX wires in as four ordered seams : ingress gate (live) → write gate → splice engine → certificate	Currently one symbol from one of thirteen files is used	—
A14b	Do not wire the write gate before NF-1/NF-3	At the measured refusal rate it would reject 83% of foreign publishes — an availability incident wearing a correctness costume	—
A15.1	Collapse dual identity: keep NextAuth + GitHub App, delete Firebase	Two auth paths is two session-fixation surfaces for one operator, plus an unused Firestore import in the client bundle	—
R-carrier	Callout > <code>[!kind]</code> for prose profiles, fenced code for opaque data. Nothing else. <code>:::</code> accepted on input, normalised away on save	An unclosed fence swallows the rest of the document (CommonMark §4.5, spec-mandated); a callout has no closing marker to lose . Measured across four engines + GitHub live. This corrects PRD §9	A carrier that beats the callout on both degradation and splice-safety. None found

1.2 Why not a CRDT — the three disqualifications

This was the single most consequential open question, and it is now closed on evidence rather than taste.

1. **A CRDT cannot own the file's bytes.** "Every untouched byte bit-identical" is unassertable across a CRDT merge, because the result derives from the operation graph, not from the bytes. Adopting a CRDT adopts precisely the architectural gap we cite against OpenKnowledge.
2. **CRDTs interleave, and interleaving is not loss the digest can see.** *The Art of the Fugue* (arXiv 2305.00583, 2023-04-30) `[fetched]`: when two users concurrently insert at the same position, the merged outcome may interleave the passages into corrupted text — *"The problem has gone unnoticed for decades, and it affects both CRDTs and Operational Transformation."* So a CRDT's convergence guarantee buys **byte-identical garbage on both devices**. Convergence is not zero loss. This is why A1b rewrites the exit condition.
3. **A CRDT cannot refuse.** Refusing is not in its algebra, and refusal is our founding principle.

And the incumbent's approach is measurably worse. `diff-match-patch@1.0.5`, executed here `[measured]`: applied twice, a patch produced `one two three four four` — and `patch_apply` returned `results = [true]` both times. It also applied an edit whose anchoring context no longer existed, returning true. Its own README states the intent: "Use best-effort to apply patch even when the underlying text doesn't match." That is architecturally the opposite of refuse-rather-than-guess. It is disqualified, not disfavoured.

Meanwhile `git`'s merge measured byte-faithful `[measured, git 2.50.1]`: preserves CRLF verbatim, preserves a missing final newline, and produces conflict markers exactly where a splice writer would want to refuse — including on a whitespace-only divergence. One mode must be banned: `--union` silently emits both sides, inventing a document with two `title:` keys.

The honest counter-argument, recorded not dismissed: conservative merging produces conflicts users hate — that is why Obsidian ships fuzzy patching. Mitigation: instrument conflict rate as a first-class shipped metric with a published budget.

1.3 Product and commercial

#	DECISION	REASONING
D15	Publishing ships in v1, but only a narrow surface: paying accounts only, <code>noindex</code> by default, system-assigned slugs, no arbitrary uploads, raw HTML and <code><form></code> stripped, hard per-account page cap	The abuse economics of a <i>paid</i> surface are inverted — the highest-ranked vectors all depend on free, automatable, indexable, vanity-named pages. A card plus <code>noindex</code> plus a random slug removes the commercial motive
	The counter-argument is about your life, not the law	The 24-hour acknowledgement clock is a permanent, unbounded, personal on-call obligation , starting the day the first stranger publishes and never ending. Missing it does not cost a fine — it collapses the s.79 safe harbour that separates "a user posted defamation" from "the founder is a co-defendant". India is the binding constraint; the EU's micro-enterprise exclusion covers us until 50 employees or €10m turnover
Pricing	₹299 and ₹599 stand. Treat ₹15,000 per transaction as an architectural constant	RBI's 2026 Framework allows recurring auth without additional-factor authentication up to ₹15,000/transaction. Software is not in the ₹1,00,000 carve-out list
	Do not offer an Indian annual plan above ₹15,000 on auto-renew	Legal, but every renewal needs on-session AFA. That is a manual repurchase wearing a subscription costume, and it reads as a billing bug
Name	<code>stetfile</code> is the strongest candidate; do not ship "Frontmatter" bare while holding qualified handles	<i>Stet</i> means "let it stand, preserve the original exactly" — literally the engine's contract. Clean on npm, GitHub, repo search and product collision. The compromise to avoid takes full trademark and SEO exposure and buys nothing. Not trademark clearance; needs counsel

1.4 Operational realities that constrain the build

FINDING	CONSEQUENCE
Cloudflare R2 has no object versioning. <code>PutBucketVersioning</code> and object-lock are absent from the S3 compatibility matrix; <code>DeleteObject</code> is a free operation; emptying a bucket is irreversible; audit logs are control-plane only, so <code>GetObject</code> / <code>PutObject</code> leave no forensic trail <code>[fetched]</code>	The standard cloud DR recipe does not port . Versioning must be built into the key layout — content-addressed immutable keys plus a small HEAD pointer — not bought. R2's own docs: "Durability does not prevent intentional or accidental deletion of data"
Indian cards get exactly one payment attempt. Smart Retries do not apply; there is a 26-hour billing delay; the bank sends a pre-transaction notice 24h ahead in its own words; the customer can opt out of any single charge; only Visa and Mastercard get mandates <code>[fetched]</code>	Involuntary churn in India has a different shape and a smaller toolkit than the global benchmarks assume
A cross-border card-not-present regime lands 2026-10-01 — 33 days out <code>[fetched, derived]</code>	If Indian customers are billed through a foreign-acquired MoR, their first charge enters that regime. Decide the rail before then
The shipped <code>globals.css</code> is byte-identical to sgnk-md's, self-labels "Linear-style modern SaaS", and carries <code>--accent: #18181b</code> with 6/8/12px radii <code>[measured]</code>	The product ships neither the canonical accent nor the square-corner rule. And a body-faint token at 2.14:1 fails WCAG AA while being in the shipped set

§2 Corrections to the PRD

These are places where later evidence overturned something the PRD asserts. **The PRD is not yet updated; treat this table as authoritative over it.**

PRD SAYS	CORRECT POSITION	SOURCE
§9: the fenced-code info string is the render dispatch mechanism	Callout for prose, fence for opaque data. An unclosed fence is document-wide; a callout has no closer	R9 m2, measured
§7.2, §24: "MDMAX is imported by zero product files"	False at import level — <code>get-snapshot.ts</code> and <code>search-index.ts</code> both import <code>decodeStrict</code> . True at capability level: 12 of 13 files have no product importer	R8 c6, measured
§7.1: mdmax has 11 tests	10 <code>*.test.ts</code> plus a fixtures directory	R8 c6, measured
Earlier session: 262 test files	98. The <code>find</code> traversed extra git worktrees	measured, twice
§24: NF-1 "recovers 99.98%"	An inference from bucketing refusal causes , not a measured result of the patched writer. Must be re-derived after the fix	R8 c6
§24 T0: exit condition is "two-device convergence"	Convergence is the wrong property (§1.2). "Zero bytes lost, every divergence surfaced as a reviewable hunk"	R11 c2, R12 k2
§3.2: "Obsidian's own community says most of a million-plus downloaders never get past their first note"	Unsourced. Traces to an unattributed pull-quote in a 2025 author blog post — no Obsidian source, no methodology, no denominator. Do not cite it	R10 s1, fetched
§19.3: frontier assumption ~\$3/\$15 per MTok	Claude Sonnet 5 is \$2/\$10 , and the scheduled increase will not occur	R7 g1, fetched
§19.3: "Gemini 2.5 Flash \$0.15/\$0.60"	That SKU does not exist. Cheapest is Gemini 3.1 Flash-Lite at \$0.25/\$1.50	R7 g1, fetched
§20.4: Dodo's flat fee is 12.7% of ₹299	That is the US rate. India-domestic is 4% + 15¢ = 4.79%. Razorpay-domestic is 2.36%, where the flat-fee argument barely applies	R7 g1, fetched
§18: the consolidation wedge is \$963/mo	\$1,163–1,223 at live prices (Mintlify moved \$250 → \$450). But the honest <i>light</i> version of the same stack is \$78 . Quote the range	R7 g1, fetched
§24: the corpus holds 7,959 frontmatter files	The pinned corpus holds 7,969 LF-terminated fences. A +10 gap, probably definitional. Reconcile before publishing either	measured here

§3 Still open — and each one re-scopes a lane

#	QUESTION	WHY IT CANNOT BE ANSWERED BY RESEARCH
D3	The name	Trademark clearance needs counsel. Everything else is decided
D4	Do documents leave the device?	Local-first collapses the DPDP/GDPR surface to near-zero and kills server-side agents, search and publishing. Server-stored enables the product and buys the full obligation set including the ₹250 crore ceiling
D9	BYO key or platform key?	Economics vs support load vs custody risk. A values call
D14	Does the agent get write authority?	It is the flagship demo and the whole lethal-trifecta liability
D17	CJK in scope for v1?	In means fixing <code>countWords</code> (1.7–2× under) and MiniSearch recall (18.1%). Out means saying so in positioning rather than shipping a silent failure
NF-4	Does <code>café</code> in NFC equal <code>café</code> in NFD for key addressing?	Both answers defensible. The one engine unit an agent cannot start
D5	The two unrotated PATs	Only you can do it. Today, not a milestone
Publishing	Will you accept a 24-hour, 365-day acknowledgement duty?	A question about your life, not about the law

§4 Research index

ROUND	AGENTS	REPORTS	ANSWERS
R1 internal	7	i1 – i7	Content engine, ecosystem inventory, knowledge-base principles, AIOS capabilities and pattern language , docs-vs-reality delta, research ledger
R2 external	8	e1 – e8	Claim verification, home surface, India market, the agent frontier , importer breakage, naming risk, fresh community signal, no protocol buyer
R3 hands-on	4	h1 – h4	The 7,959-file foreign-corpus run , the Hubble/Front Matter CMS teardowns, the OpenKnowledge teardown
R4 editors	5	ed1 – ed5	Writing tools, IDEs, lossy frameworks , AI editors, the office-suite review-loop canon
R5 AI journey	4	aj1 – aj4	Agent protocols, chat-to-artifact laws behind <code>land()</code> , knowledge formats, the journey-home gap
R6 core	6	c1 – c6	The projection law , the computation budget, the B2B wedge, INR pricing, build refs, twelve screens
R7 PRD corpus	16	x1 – x13 , g1 – g3	Lossless compression of R1–R6 + business model and unit economics , internal systems fusion , risk/legal/security/capacity
R8 AIOS + specDD	12	a1 – a5 , b1 – b5 , c5 , c6	AIOS tools and loops as product, AIOS as a B2B surface, SKILL.md as user automation, markdown processing assets, the spec-driven-development market , the document-type canon, prompts as documents, the AI build loop , spec-conformance verification, the spec-file system , build-plan inputs
R9 markdown	10	m1 – m10	Spec landscape, the carrier decision , the extension catalogue, frontmatter key conventions, typed markdown, render pipelines, output targets, computational markdown, hard edges, the render possibility space
R10 untouched angles	7	s1 – s4 , d1 , d2 , d4	Simplicity engineering , editor latency, the affordance problem, mobile, search, collaboration models, AI document interaction
R11 gaps	11	c1 – c4 , d3 , r1 – r6	The research gap audit , architecture decisions , feature completeness, onboarding, longevity, migration fidelity, the academic segment, AI disclosure, naming candidates , measurement, support economics
R12 build blockers	14	k1 – k14	DR (R2 has no versioning) , the sync engine , refusal UX, accessibility, API design, observability, roles, abuse and publishing , desktop distribution, performance and testing, data model and migrations, i18n/RTL/IME, OSS/docs/community, Indian billing ops

Paths: docs/research/agent-reports-2026-08-28/ (R1–R6) · -r7/ · -r8to10/ · -r11/ · -r12/ .

§5 What we may say in public, and what we may not

May say — each is `[measured]` or `[fetched]` and re-derivable:

- 0 corruption and 0 throws across 8,513 third-party markdown files from 7 vaults, byte-pinned and re-verifiable by anyone via `npm run corpus`.
- Three competitors' write paths executed, with the exact bytes each destroys.
- Every mainstream rich-text framework is lossy by design, with the vendors' own words for it.

May not say, yet:

- **Any fidelity or coverage percentage.** At the measured 83% refusal rate the headline number is false until NF-1 and NF-3 land. Publishing first converts a bug into a public claim.
- **"NF-1 recovers 99.98%."** It is an inference, not a measurement.
- **Anything about Obsidian's onboarding funnel.** The widely-repeated claim is unsourced.
- **"First AI attribution."** Cursor and Grammarly exist. The defensible claim is narrower: the first markdown editor with byte-anchored, document-portable, reader-visible provenance.
- **"Serve markdown to agents and get cited."** Measurably refuted.
- **Any `[SS]`-tagged number.** 52 of them remain in the PRD.

Three documents, one system: the PRD says what to build, the build plan says in what order and how it is proven, and this says what was decided and what turned out to be wrong. When they disagree, this one is newer.